

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech CST 3rd Semester Final Examinations, November 2024

Data Structures CS-2103

Full Marks: 50

Time: 3 hours

Attempt all questions

1. [Mandatory question] Answer all of the following questions

- a) Write a C function to count the total number of nodes of a *Perfect Binary Tree* in an efficient way.

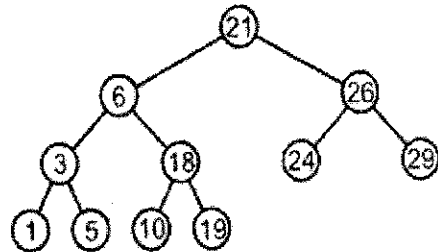
Function prototype looks as follows - `int nodeCount(treeNode * root);`
where `treeNode` is declared as linked-node approach as below -

```
typedef struct treeNode {  
    int info; // information field  
    struct treeNode *lchild; // left child  
    struct treeNode *rchild; // right child  
} treeNode
```

- b) Consider the following AVL tree with eleven nodes shown in the diagram.

Show the new AVL tree upon inserting an additional node with value 20.

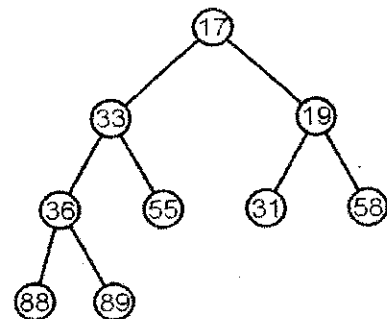
Provide detail steps and justification for your answer.



- c) Consider the min-heap with nine nodes shown in the diagram. Do the following with brief explanation -

(i) Show the resulting min-heap after inserting an element with value 25.

(ii) Show the resulting min-heap on performing delete operation (removing the root) *once* from the resulting heap in step (i).



$$(5+5+6 = 16)$$

2. Answer **any 6 (not more than 6)** of the following questions

- a) Write down the following
 - i. The postfix form of the infix expression: $D * (A + B) + E / (F + A * G) + C$
 - ii. The infix form of the postfix expression: $A B + C D * -$
- b) What is a Binary Search Tree? What would be the worst case search time in a binary search tree having n nodes in it?
- c) Suppose the running time analysis of a particular algorithm results in $(2n + 5)$ primitive operations. Explain why complexity of the algorithm is $O(n)$?
- d) A Binary Search Tree (BST) stores integer elements and is implemented using 1-D array-based approach. Write a C function to insert an element in the BST.
- e) Explain different methods for resolving collision in hashing and mention their advantages and disadvantages.
- f) Prove that a *Perfect Binary Tree* of height h has 2^h leaf nodes.
- g) Write down a C function to check whether a given binary tree is full (strict) or not
- h) Mention properties of good hash function in the context of hashing. How do you map a hash value to respective hash buckets?
- i) Write a short note on "Threaded Binary Tree".

(6x4 = 24)

3. Answer **any 2 (not more than 2)** of the following questions using 'C' language

- a) Write the structure for array-based Queue that stores integers. Write functions for the following operations - initialize the queue, enqueue, and dequeue.
- b) A binary tree stores integer elements and is implemented using the linked-node representation. Write a C function to print and count the total number of full nodes (nodes with precisely two children).
- c) Given a singly linked list containing integer data. Write a function that adds all nodes' data and stores the sum as a new node at the last of the list. For example, if the given linked list is $6 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow \text{NULL}$, then your function should modify the linked list to $6 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 14 \rightarrow \text{NULL}$.

(2x5 = 10)